SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Concept Mapping

Code of Conduct:

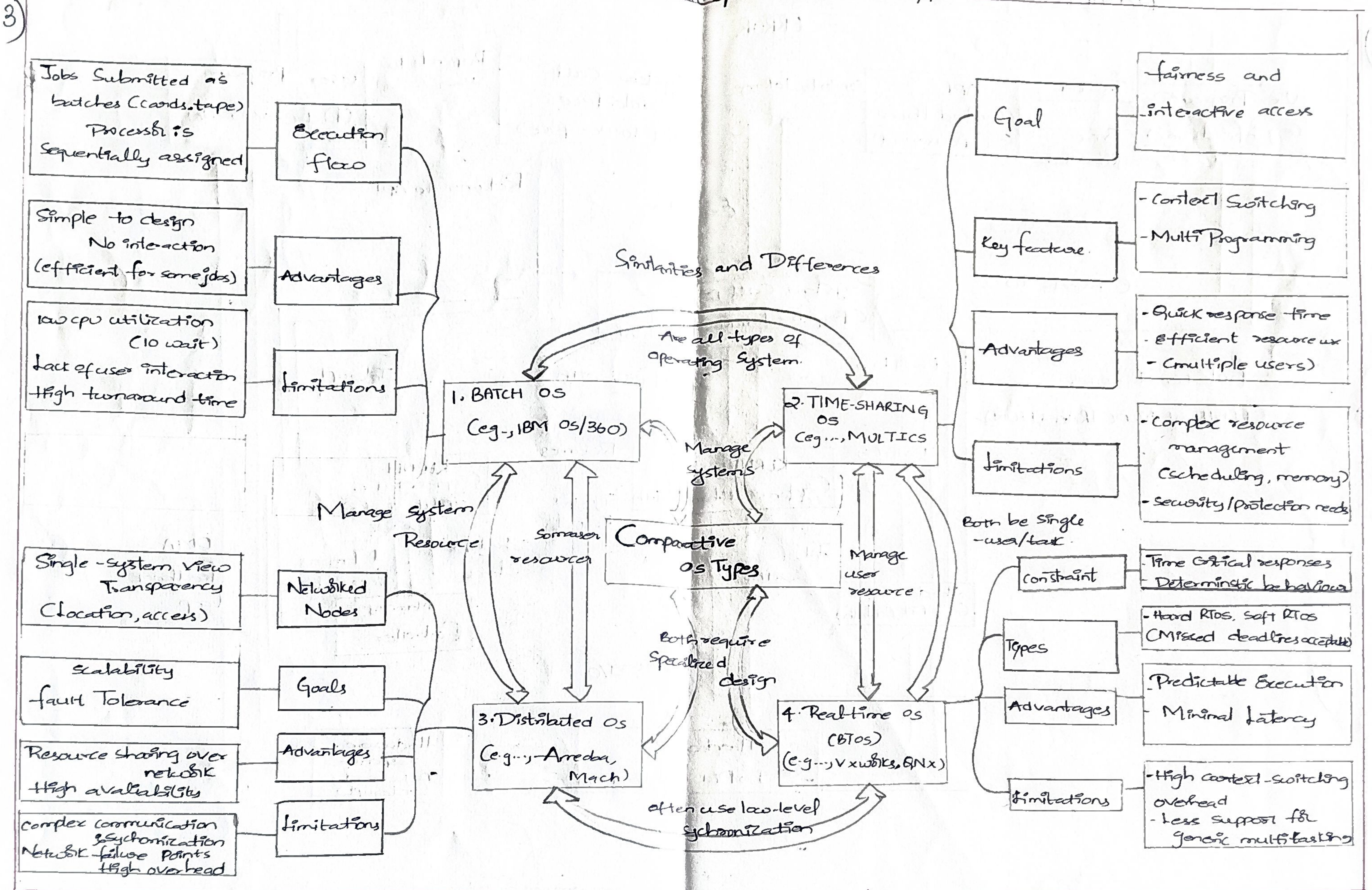
S. Naukesh (192524559) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Naukesh

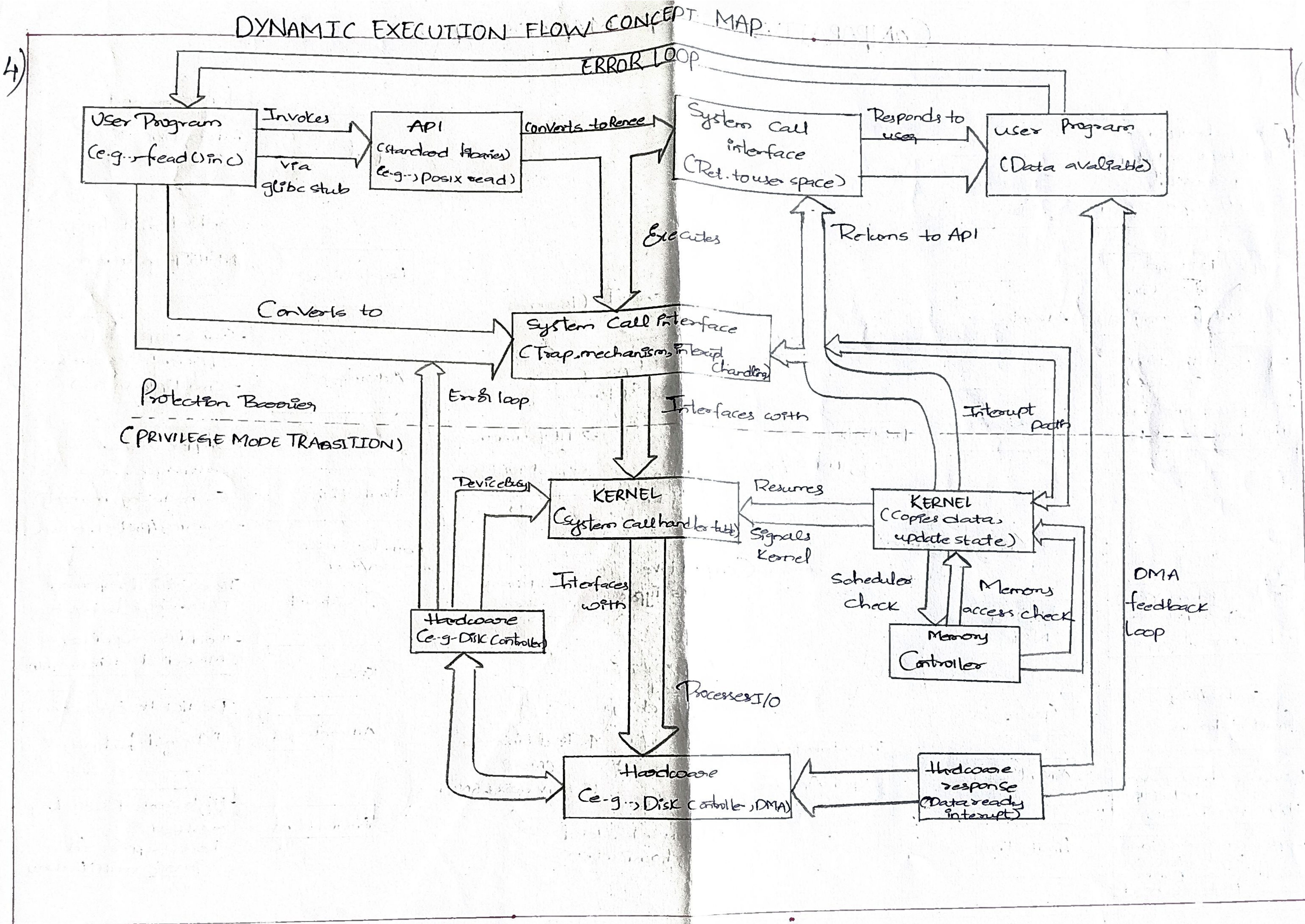
Signature of the student:

3)

COMPARATIVE CONCEPT MAP: OS TYPES



4)


DYNAMIC EXECUTION FLOW/ CONCEPT MAP
ERROR LOOP
User Program
(e.g., read() in C)
Invokes
via
glibc stub
API
(standard libraries)
(e.g., POSIX read)
Converts to Renee
System Call interface
(Ret. to user space)
Responds to user
User Program
(Data available)
Executes
Returns to API
Converts to
System Call Interface
(Trap mechanism, Interrupt handling)
Protection Barrier
(PRIVILEGE MODE TRANSITION)
Error loop
Interfaces with
Interrupt path
Device Busy
KERNEL
(System call handler table)
Resumes
Signals Kernel
KERNEL
(Copies data, update state)
Interfaces with
Scheduler check
Memory access check
Hardware
(e.g. Disk Controller)
Memory Controller
DMA feedback loop
Processes I/O
Hardware
(e.g., Disk controller, DMA)
Hardware response
(Data ready interrupt)